Spin–Momentum Filtering by an Absorbing Boundary Delays Quantum Detection in a Harmonic Waveguide

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Absorbing boundaries are often treated as scalar sinks. We show that a spin-coupled boundary for a Pauli particle acts instead as a spin–momentum impedance. For each tangential mode $\boldsymbol{\xi}$, the boundary matrix has eigenbranches $i\kappa \pm |\boldsymbol{\xi}|$ at fixed κ; in a harmonic guide $|\boldsymbol{\xi}| \sim \sqrt{\omega}$. Narrowing the guide (larger ω) strengthens an evanescent boundary response without a bulk potential barrier, suppresses prompt roof flux, lowers the fixed-window detected fraction, and shifts the detector-present response to later detection times; over that window, the restricted mean detection time is fitted by $A + B\sqrt{\omega}$.

Introduction— Absorbing boundaries are often introduced to remove probability from an open quantum evolution. In detector models this loss is physical: the squared norm $\|\Psi_t\|^2$ is the probability that no click has yet occurred, and its loss rate is the click-time density [1–3]. The boundary is therefore not merely a surface where probability disappears. Like an impedance in wave physics, it is an interface law determining which components are absorbed, reflected, mixed, or filtered. Its response is part of the measurement, not only a numerical device [4, 5].

This Letter studies a spin-coupled absorbing boundary condition (spinor ABC) whose impedance is matrix-valued. For a nonrelativistic spin-1/2 particle in a harmonic waveguide, the absorbing boundary couples the normal derivative of the spinor to its tangential derivatives. Tangential Fourier decomposition at the roof turns the boundary condition into a 2×2 matrix relation $\partial_z \widehat{\Psi} = \mathcal{C}(\boldsymbol{\xi})\widehat{\Psi}$. The eigenvalues $i\kappa \pm |\boldsymbol{\xi}|$ of $\mathcal{C}(\boldsymbol{\xi})$ define two spin–momentum branches. For the harmonic transverse ground state, $\ell_\perp = \omega^{-1/2}$ in dimensionless units; hence increasing ω narrows the guide and raises the typical tangential scale sampled at the roof, so that $|\boldsymbol{\xi}| \sim \ell_\perp^{-1} \sim \sqrt{\omega}$. This local boundary scale underlies the delayed detector-present roof-flux response studied here.

This response is detector-present. The observable is the roof flux, equivalently the norm loss, of a specified nonunitary absorbing-boundary evolution. It is not the no-detector Bohmian first-arrival distribution of a freely evolving Pauli wave [6–8]. Nor is the spin-coupled absorbing boundary assumed to describe all possible detecting screens; it is a particular idealized hard-detector model, a restriction emphasized by recent scattering-theory comparisons of absorbing detectors [9].

The Das–Dürr waveguide geometry provides a sharp comparison between detector-free arrival and detector-present detection [7]. In the same separable guide, complex absorbing potentials and the spin-decoupled absorbing boundary behave as scalar absorbers: they may generate reflected tails through detector back-action, but they do not read transverse confinement in this way [10, 11]. By contrast, the spin-coupled absorbing boundary suppresses the prompt roof-flux peak, lowers the detected fraction in a fixed observation window, and converts the transverse confinement into a delayed oscillatory sector, while showing no appreciable dependence on the initial Bloch angles in the tested regimes [11, 12]. Recent simulations revealed an unexplained confinement dependence for that spin-coupled absorbing boundary [11]. Here we identify its local origin, derive the two-branch boundary symbol $i\kappa \pm |\boldsymbol{\xi}|$, and show how it produces a distinctive ω-dependent detector-response signature.

We work in units $\hbar = m = 1$. The bulk Hamiltonian is spin independent,

$$H = -\frac{1}{2}\Delta + \frac{1}{2}\omega^2\left[x^2 + y^2\right], \tag{1}$$

and the initial state is factorized as

$$\Psi_0(x, y, z) = \chi_\omega(x, y)\,\phi_0(z)\,\eta, \tag{2}$$

where $\eta \in \mathbb{C}^2$ is a constant spinor and χ_ω is the transverse harmonic ground state. For the finite-window confinement study below, ϕ_0 is a right-moving Gaussian longitudinal packet centered midway along the guide. The finite guide has a reflecting lower end at $z = 0$, and the detecting surface is the roof

$$\Sigma_L = \{z = L\}. \tag{3}$$

On this surface we impose the spin-coupled absorbing boundary condition

$$(\boldsymbol{\sigma} \cdot \boldsymbol{\nabla})\Psi = i\kappa\sigma_z\Psi, \qquad \kappa > 0. \tag{4}$$

This is the nonrelativistic limit of the semi-ideal Dirac absorbing boundary [13]; well-posedness and contraction-semigroup formulations are discussed in Refs. [14, 15]. For the Pauli current

$$\boldsymbol{j}^P = \mathrm{Im}(\Psi^\dagger\boldsymbol{\nabla}\Psi) + \frac{1}{2}\boldsymbol{\nabla} \times (\Psi^\dagger\boldsymbol{\sigma}\Psi), \tag{5}$$

the spinor ABC gives the outward roof flux

$$j_z^P(x,y,L,t) = \kappa\rho(x,y,L,t), \qquad \rho = \Psi^\dagger\Psi. \tag{6}$$

By the continuity equation, with no flux through the remaining faces, the detector-present click-time density is

$$g(t;\omega) = \kappa \int_{\Sigma_L} \rho(x,y,L,t)\, dxdy = -\frac{d}{dt}\|\Psi_t\|^2. \tag{7}$$

For a finite observation horizon T, define the detected fraction

$$D_T(\omega) = \int_0^T g(t;\omega)\, dt, \tag{8}$$

and the finite-window restricted mean detection time,

$$\mu^*(T;\omega) = \int_0^T S(t;\omega)\, dt, \qquad S(t;\omega) = \|\Psi_t\|^2, \tag{9}$$

or,

$$\mu^*(T;\omega) = \mathbb{E}[\min(\tau, T)]. \tag{10}$$

Here τ is the random detector click-time. We use this restricted mean detection time because, at strong confinement, a substantial part of the probability remains undetected at the end of the finite observation window. Bohmian histograms shown in the plots are only Monte Carlo samples of the same detector-present flux law. These trajectories obey

$$\dot{\boldsymbol{Q}}(t) = \frac{\boldsymbol{j}^P[\Psi_t^{\mathrm{ABC}}](\boldsymbol{Q}(t))}{\rho[\Psi_t^{\mathrm{ABC}}](\boldsymbol{Q}(t))}. \tag{11}$$

Here $\boldsymbol{Q}(t)$ denotes a sampled Pauli-current trajectory and Ψ_t^{ABC} denotes the nonunitary spinor-ABC evolution.

We now present the confinement-dependent roof flux and then derive the boundary-symbol mechanism responsible for the $\sqrt{\omega}$ scale.

Confinement-dependent detection— Figure 1 shows the central numerical observation. The red curve in Fig. 1 is the detector-free one-dimensional Gaussian current through $z = L$ for the same longitudinal packet, with the transverse factor integrated out. It is not a detector model or fit input; it only marks the free crossing scale defined in detail in Ref. [11]. Only the transverse confinement parameter ω is varied; the longitudinal packet, detector parameter, initial spinor, and observation window are fixed. Increasing ω narrows the transverse ground state and strongly suppresses the prompt roof-flux peak. This prompt suppression is the most direct finite-window signature of the boundary response. The detected fraction drops from $D_{20} \simeq 0.93$ at $\omega = 1$ to $D_{20} \simeq 0.42$ at $\omega = 300$. Figure 1(c) summarizes the same confinement sweep: over the sampled ω-values, the restricted mean detection time is fitted by

$$\mu^*(20;\omega) \simeq 4.084 + 0.638\sqrt{\omega}. \tag{12}$$

The fit in Eq. (12) is a finite-window observation, not an asymptotic law. Its coefficients would depend on the prepared wave packet, detector parameter, guide geometry, and observation time. The robust feature is the square-root scale. We now identify its boundary origin and the finite-guide mechanism that makes this scale visible in the roof-flux detection time statistic.

Boundary mechanism and delayed return— We discuss the mechanism through three logical steps: (i) ordinary first-pass bulk propagation is excluded as the source of the ω-dependence; (ii) the spinor ABC is shown to introduce the local roof scale $R = |\boldsymbol{\xi}|$; and (iii) finite-guide memory provides a natural route by which undetected, branch-reweighted components can return to the roof and contribute to the delayed detection-time sector in Fig. 1.

(i) The square-root scale is not generated by ordinary first-pass bulk propagation. Before significant feedback from the detecting roof, the spin-independent separable bulk evolution has the first-pass form

$$\Psi_{\mathrm{fp}}(x,y,z,t) = \chi_\omega(x,y,t)\phi_{\mathrm{fp}}(z,t)\eta, \tag{13}$$

where $\phi_{\mathrm{fp}}(z,0) = \phi_0(z)$. The transverse harmonic ground state changes only by an overall phase, so its density is unchanged, and the spinor η remains constant because the bulk Hamiltonian is spin independent. For this first-pass field, the Pauli current can then be written

$$\boldsymbol{j}_{\mathrm{fp}}^P = \boldsymbol{j}_{\mathrm{fp}}^{\mathrm{conv}} + \frac{1}{2}\boldsymbol{\nabla}\times(\rho_{\mathrm{fp}}\boldsymbol{s}_\eta), \quad \boldsymbol{s}_\eta := \eta^\dagger\boldsymbol{\sigma}\eta, \quad \rho_{\mathrm{fp}} := |\Psi_{\mathrm{fp}}|^2. \tag{14}$$

Since $\boldsymbol{s}_\eta$ is constant in the first-pass separable approximation, the z-component of the spin-curl term is a transverse divergence. By Stokes' theorem on the transverse cross-section, together with the lateral Dirichlet conditions, its cross-section integral vanishes. Hence, for an interior plane Σ_b below the roof,

$$\int_{\Sigma_b} j_{\mathrm{fp},z}^P\, dA = \int_{\Sigma_b} \mathrm{Im}(\Psi_{\mathrm{fp}}^\dagger \partial_z \Psi_{\mathrm{fp}})\, dA = \mathrm{Im}(\phi_{\mathrm{fp}}^* \partial_z \phi_{\mathrm{fp}})(z_b, t). \tag{15}$$

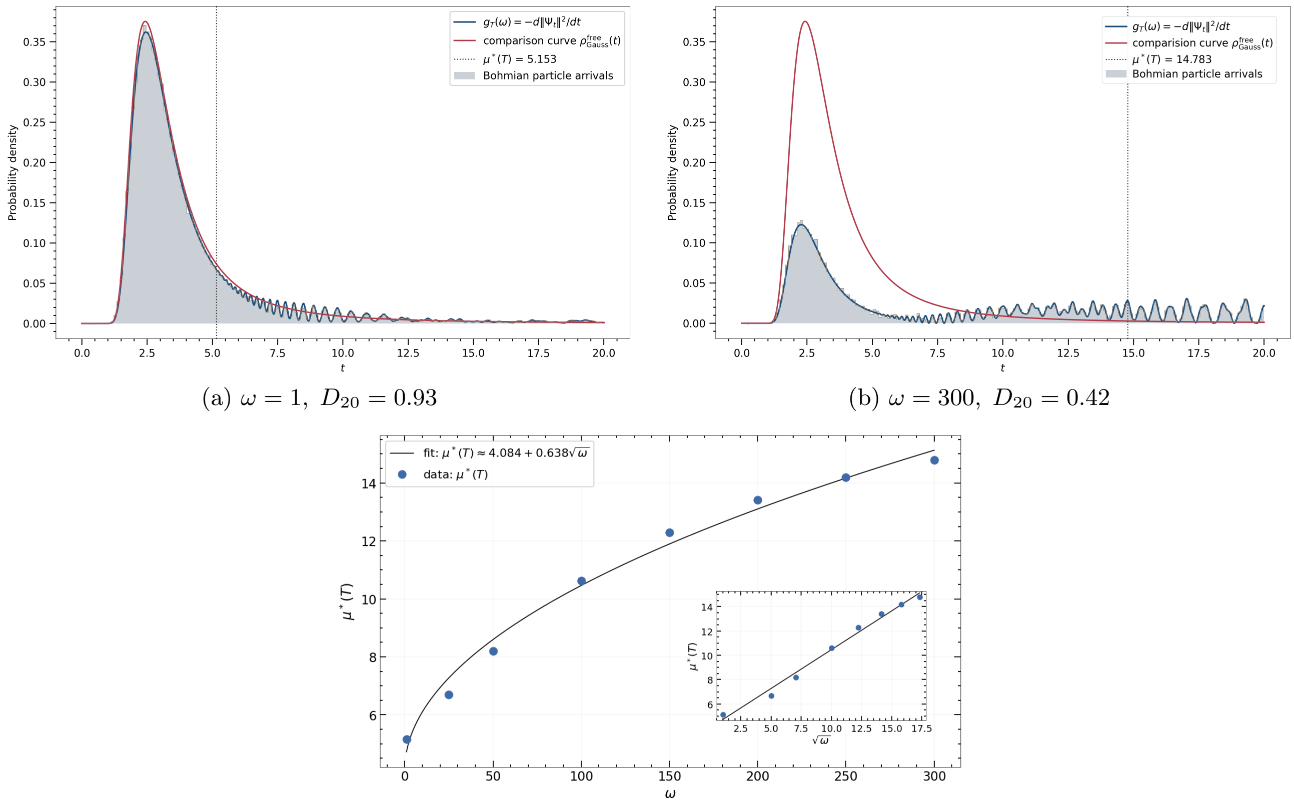
The leading plane-integrated first-pass flux is therefore independent of the transverse confinement parameter ω. More details on the first-pass bulk calculation are given in the Supplemental Material [16], Sec. S5.

(ii) The observed confinement dependence therefore must enter when the wave interacts with the detecting boundary. The spinor ABC introduces this scale through the tangential symbol $\mathcal{C}(\boldsymbol{\xi})$ of the boundary operator. The same roof condition, Eq. (4), can be written componentwise as

$$\begin{aligned} \partial_z\psi_\uparrow &= i\kappa\,\psi_\uparrow - (\partial_x - i\partial_y)\psi_\downarrow, \\ \partial_z\psi_\downarrow &= i\kappa\,\psi_\downarrow + (\partial_x + i\partial_y)\psi_\uparrow. \end{aligned} \tag{16}$$

Equivalently,

$$\partial_z\Psi = \mathcal{C}\Psi, \quad \mathcal{C} = \begin{pmatrix} i\kappa & -D_- \\ D_+ & i\kappa \end{pmatrix}, \quad D_\pm = \partial_x \pm i\partial_y. \tag{17}$$



(a) $\omega = 1,\ D_{20} = 0.93$

(b) $\omega = 300,\ D_{20} = 0.42$

(c) restricted mean detection time, $T = 20$

FIG. 1. Confinement-induced delay for the spin-coupled ABC. (a),(b) Detector-present roof-flux density $g(t;\omega) = -d\|\Psi_t\|^2/dt$ for $\omega = 1$ and $\omega = 300$. red: detector-free Gaussian comparison current; blue: spinor-ABC roof flux; gray histograms: trajectory samples of the same spinor-ABC roof-flux law, not detector-free Bohmian arrival times. Only ω is varied; $\kappa = k_z = \pi$, $\sigma_\parallel = 0.5$, and $T = 20$ are fixed. Increasing confinement suppresses the prompt peak and lowers D_{20} from 0.93 to 0.42. (c) Restricted mean detection time $\mu^*(T;\omega)$, fitted by $\mu^*(20;\omega) \simeq 4.084 + 0.638\sqrt{\omega}$ for several ω. Details of the numerical setup, full confinement sweep, finite-window bookkeeping, and convergence tests are given in the Supplemental Material [16], Secs. S2–S4.

Therefore the detector couples normal absorption to tangential derivatives. For a tangential Fourier mode $\boldsymbol{\xi} = (\xi_x, \xi_y)$, Eq. (17) gives

$$\mathcal{C}(\boldsymbol{\xi}) = i\kappa I + (\hat{z} \times \boldsymbol{\xi}) \cdot \boldsymbol{\sigma}. \tag{18}$$

Writing

$$R := |\boldsymbol{\xi}|, \qquad \Gamma(\boldsymbol{\xi}) = \frac{(\hat{z} \times \boldsymbol{\xi}) \cdot \sigma}{R}, \qquad \Gamma^2 = I. \tag{19}$$

Thus

$$\mathcal{C}(\boldsymbol{\xi}) = i\kappa I + R\Gamma(\boldsymbol{\xi}). \tag{20}$$

The two tangential spin–momentum projectors are then

$$\Pi_\pm(\boldsymbol{\xi}) = \frac{1}{2}\big(I \pm \Gamma(\boldsymbol{\xi})\big), \tag{21}$$

and the corresponding boundary eigenbranches are

$$\lambda_\pm(\boldsymbol{\xi}) = i\kappa \pm R. \tag{22}$$

This is the key point: the detector imposes not only the impedance scale κ, but also the tangential spin–momentum scale $R = |\boldsymbol{\xi}|$. The same branch structure gives the local normal response. Let $W(\boldsymbol{\xi}, t) = \widehat{\Psi}(\boldsymbol{\xi}, L, t)$ be the roof trace. For a small auxiliary inward depth $\epsilon = L - z$, the homogeneous normal continuation generated by the boundary relation is

$$\widehat{\Psi}(\boldsymbol{\xi}, L - \epsilon, t) = e^{-i\kappa\epsilon} B_{\mathrm{br}}(\epsilon, \boldsymbol{\xi}) W(\boldsymbol{\xi}, t) + \widehat{\mathcal{R}}_{\mathrm{Duh}}, \tag{23}$$

where

$$B_{\mathrm{br}}(\epsilon, \boldsymbol{\xi}) W = e^{-R\epsilon}\Pi_+ W + e^{R\epsilon}\Pi_- W. \tag{24}$$

This is a matrix-valued branch filter acting on the spinor amplitude $W(\boldsymbol{\xi}, t)$: it decomposes the roof trace into the two tangential spin–momentum components $\Pi_\pm W$ and assigns them the auxiliary normal weights $e^{-R\epsilon}$ and $e^{R\epsilon}$. The term $\widehat{\mathcal{R}}_{\mathrm{Duh}}$ is a Duhamel remainder measuring the deviation of the true time dependent Schrödinger equation (TDSE) solution from this homogeneous boundary continuation away from the roof; it has no zeroth- or first-order contribution at $\epsilon = 0$, as shown in the Supplemental Material [16], Sec. S7. All physical detection statistics remain defined at the roof, $\epsilon = 0$. Therefore

$B_{\rm br}$ is not a physical propagation law through a detector layer, but a local boundary-symbol probe

$$B_{\rm br}(0,\boldsymbol{\xi}) = I, \qquad \partial_\epsilon B_{\rm br}(\epsilon,\boldsymbol{\xi})|_{\epsilon=0} = -R\,\Gamma(\boldsymbol{\xi}). \tag{25}$$

The branch structure is not a direct multiplicative suppression of the roof density; it enters the first inward normal response. Therefore the same tangential radius $R = |\boldsymbol{\xi}|$ that labels the roof mode controls the local normal boundary response. For the harmonic transverse ground state, $\ell_\perp = \omega^{-1/2}$, so the typical tangential Fourier scale is set by

$$R = |\boldsymbol{\xi}| \sim \ell_\perp^{-1} \sim \sqrt{\omega}. \tag{26}$$

This is the local boundary scale. To express how it enters a normalized first time moment, define an auxiliary boundary layer response measure

$$d\mu^{\rm bl}_{\omega,\epsilon}(\boldsymbol{\xi},t) := |B_{\rm br}(\epsilon,\boldsymbol{\xi})W(\boldsymbol{\xi},t)|^2\, d^2\xi\, dt. \tag{27}$$

This is not a click-time distribution at $z = L-\epsilon$: the variable t is only the time label of the roof trace $W(\boldsymbol{\xi},t)$, and the physical detector observable remains the roof flux at $z = L$. Since the scalar phase drops out of densities and the Duhamel term has no first-order contribution at the roof, the first variation is computed from the homogeneous branch probe. With $B_{\rm br} = e^{-\epsilon R\Gamma}$,

$$|B_{\rm br}W|^2 = W^\dagger e^{-2\epsilon R\Gamma} W = |W|^2 b_{\omega,\epsilon}. \tag{28}$$

where

$$b_{\omega,0} = 1, \qquad \partial_\epsilon b_{\omega,\epsilon}|_{\epsilon=0} = R a_\omega, \tag{29}$$

and

$$a_\omega(\boldsymbol{\xi},t) := -2\frac{W^\dagger\Gamma(\boldsymbol{\xi})W}{W^\dagger W}, \qquad |a_\omega| \le 2. \tag{30}$$

To normalize the auxiliary boundary-response measure, let $d\nu^{(0)}_\omega = d\mu^{\rm bl}_{\omega,0}/\int d\mu^{\rm bl}_{\omega,0}$. After normalization,

$$\mathbb{E}_{\nu^{\rm bl}_{\omega,\epsilon}}[F] = \frac{\mathbb{E}_{\nu^{(0)}_\omega}[F b_{\omega,\epsilon}]}{\mathbb{E}_{\nu^{(0)}_\omega}[b_{\omega,\epsilon}]}. \tag{31}$$

Differentiating at $\epsilon = 0$ gives

$$\partial_\epsilon \mathbb{E}_{\nu^{\rm bl}_{\omega,\epsilon}}[F]\big|_{\epsilon=0} = \mathrm{Cov}_{\nu^{(0)}_\omega}(F, R a_\omega). \tag{32}$$

Take $F = t$, the local first-order boundary sensitivity is

$$\Lambda_\omega = \mathrm{Cov}_{\nu^{(0)}_\omega}(t, R a_\omega). \tag{33}$$

For the harmonic transverse family, $R = \sqrt{\omega}s$, with s dimensionless and order one. Hence

$$\Lambda_\omega = \sqrt{\omega}\;\mathrm{Cov}_{\nu^{(0)}_\omega}(t, s a_\omega) =: \sqrt{\omega}\,\beta_\omega. \tag{34}$$

Thus Λ_ω is a local first-order boundary sensitivity, not the observed finite-window delay itself. Equation (34) shows that the first boundary-response contribution factorizes into the explicit transverse scale $\sqrt{\omega}$ and a dimensionless roof-trace covariance β_ω. The latter is not universal: it depends on the detector-present roof trace and may change with the prepared wave packet, detector parameter, guide geometry, and observation window. The finite-grid diagnostics in the Supplemental Material [16], Sec. S7, show $\beta_\omega = O(1)$ over the sweep, with no competing power-law growth. Thus the local calculation identifies the $\sqrt{\omega}$ scale, not the setup-dependent value or sign of B in the finite-window fit $A + B\sqrt{\omega}$.

(iii) In the finite guide, the part of the wave not removed during the first near-roof interaction is not a scalar reflected copy of the incident packet. It has already been attenuated by detector loss and branch-reweighted by the spinor ABC. Subsequent evolution can bring this surviving, filtered amplitude back to the roof again, where it contributes to the delayed oscillatory detection sector. The Supplemental Material [16], Sec. S8, formulates this as a reduced flux-level bookkeeping closure with an effective memory kernel, not as an exact mode-resolved TDSE decomposition.

Conclusion— We have shown that the spinor ABC is not a scalar absorbing sink but a local spin–momentum impedance. At the detecting roof, each tangential mode decomposes into two spin–momentum components $\Pi_\pm W$, with boundary eigenvalues $i\kappa \pm |\boldsymbol{\xi}|$. In a harmonic guide the roof trace samples $R = |\boldsymbol{\xi}| \sim \ell_\perp^{-1} \sim \sqrt{\omega}$, so increasing transverse confinement directly changes the local detector-present boundary response. Equation (24) gives the local meaning of this filtering: after the scalar phase is removed, the auxiliary normal response weights the two spin–momentum components $\Pi_\pm W$ by $e^{\mp R\epsilon}$. The $e^{-R\epsilon}$ factor is an evanescent boundary response, not produced by a bulk potential barrier, and the companion $e^{R\epsilon}\Pi_-$ branch is not a gain channel. The observable statistic remains the roof flux at $z = L$.

Together with finite-guide memory of the undetected, branch-reweighted amplitude, this local scale organizes the prompt-flux suppression, the reduction of D_{20}, and the growth of the finite-window mean in Fig. 1. Accordingly, Eq. (12) should be read as a finite-window fit for the specified prepared wave packet, detector parameter, guide geometry, and observation time. The constants A and B are effective and setup dependent. The robust observable signature is the confinement-controlled deformation of the detector-present roof flux, not the numerical value of the fitted coefficient.

A direct physical realization of this effect would require engineering an absorbing interface whose effective entrance impedance realizes the tangential branches $i\kappa \pm |\boldsymbol{\xi}|$, rather than relying on generic spin-dependent loss or bulk spin–orbit coupling; controls, parameter scans, and physical scales are given in the Supplemental Material [16], Secs. S9–S11.

Code and data availability. Code, summary data, and representative density animations are available in the repository `Boundary-Spin-Momentum-Filtering-Data`. The solver follows the Crank–Nicolson/GMRES implementation described in Appendix A of Ref. [11].

Declarations. OpenAI ChatGPT was used for language polishing and for assistance with Python scripting, debugging, and post-processing. The author reviewed all outputs and takes full responsibility for the manuscript, analysis, code, results, and conclusions.

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[16] See Supplemental Material at [URL will be inserted by publisher] for numerical details, convergence tests, boundary-symbol and Duhamel derivations, reduced return bookkeeping, controls, parameter scans, and physical scale conversions.